Day 100 | Unit 07 | Teacher plan

Understand and Use Percent

Explain the Reference Whole in Percent Models

Standards: 6.RP.A.3c

Learning target: I can explain the Reference Whole in Percent Models, and explain and check my reasoning.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer. Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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01 / Learn

Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer. Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole.

Worked example 1

50% of a number is 60. Find the number and check.

1. Write part = decimal rate × whole.

2. Divide the part by the decimal rate to find the whole.

3. Whole = 60 ÷ 1/2 = 120.

4. Check 50% × 120 = 60.

5. State the result in the form requested: 120.

Answer: 120

Worked example 2

A school club has 80 members; 30% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

1. Use part = (percent/100) × whole.

2. Rename 30% as 3/10.

3. Multiply by 80 to obtain 24.

4. Compare the rate with 1.

Answer: 24 participants; n = (30/100) × 80; less than the whole because the percent is below 100

Day 100 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

25% of a number is 45. Find the number and check.

02 / Constructed response

Find 25% of 220.

03 / Constructed response

25% of a number is 35. Find the number and check.

04 / Constructed response

Find 15% of 120.

05 / Constructed response

125% of a number is 50. Find the number and check.

06 / Constructed response

Find 40% of 220.

07 / Constructed response

11 students are 5% of a school’s enrollment. Find the enrollment.

08 / Numeric entry and equation

A school club has 100 members; 25% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

09 / Multiple select

24 is 20% of an unknown quantity w. Select all equations that find w.
A. $w = 24 \times (20/100)$
B. $w = 24 + 20$
C. $(20/100)w = 24$
D. $w = 24 \div (20/100)$

10 / Constructed response

Find 12% of 180. Then explain a mental-math or fraction strategy that checks your answer.

Day 100 | Unit 07 | Answer key

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1. 180

Whole = 45 ÷ 1/4 = 180. Check 25% × 180 = 45.

6.RP.A.3c | Constructed response

2. 55

25/100 × 220 = 55.

6.RP.A.3c | Constructed response

3. 140

Whole = 35 ÷ 1/4 = 140. Check 25% × 140 = 35.

6.RP.A.3c | Constructed response

4. 18

15/100 × 120 = 18.

6.RP.A.3c | Constructed response

5. 40

Whole = 50 ÷ 5/4 = 40. Check 125% × 40 = 50.

6.RP.A.3c | Constructed response

6. 88

40/100 × 220 = 88.

6.RP.A.3c | Constructed response

7. 220

Whole = 11 ÷ 1/20 = 220. Check 5% × 220 = 11.

6.RP.A.3c | Constructed response

8. 25 participants; n = (25/100) × 100; less than the whole because the percent is below 100

Rename 25% as 1/4. Multiply by 100 to obtain 25. Compare the rate with 1.

6.RP.A.3c | Numeric entry and equation

9. C, D - (20/100)w = 24; w = 24 ÷ (20/100)

The part is percent rate times whole. Divide 24 by 1/5; w = 120. Check by multiplication.

6.RP.A.3c | Multiple select

10. 108/5

12/100 × 180 = 108/5.

6.RP.A.3c | Constructed response